

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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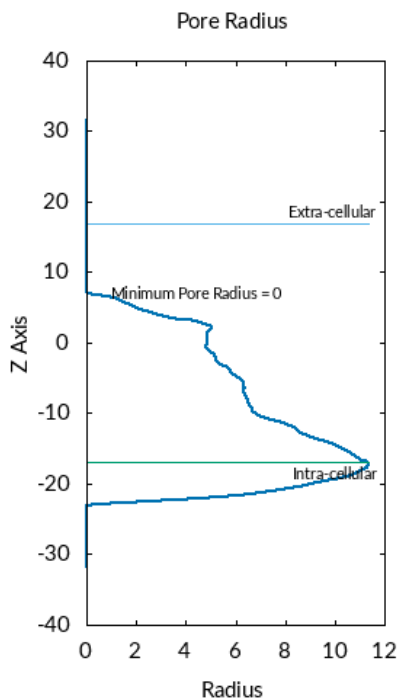
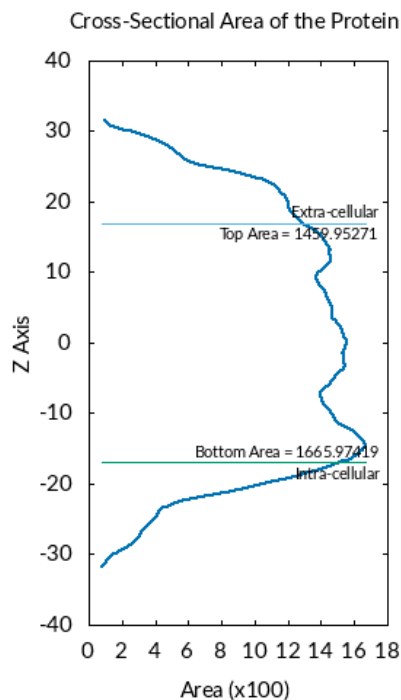
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PDB Info STEP 1 STEP 2 STEP 3 STEP 4 STEP 5 STEP 6 JOB ID: 3750189754

CHARMM PDB: [step1_pdbreader.pdb](#) ([view structure](#))
 Orientation Input: [step2_orient.inp](#)
 Orientation Output: [step2_orient.out](#)
 Oriented PDB: [step2_orient.pdb](#) ([view structure](#)) (please view this structure before you move to the next step)
 Area Calculation: [step2_area.str](#) Calculate cross sectional area of the protein
[step2_area.plo](#) Computed cross sectional area along Z axis
[step2_protein_area.str](#) Top/Bottom area of the protein
 Pore Water: [step2.1_pore.inp](#)
[step2.1_pore.out](#)
[step2.1_pore.pdb](#) ([view structure](#))
[step2.1_porewat.pdb](#) ([view structure](#))
[step2.1_porewat.crd](#)

[download.tgz](#)

Calculated Cross Sectional Area:



Protein Top Area: 1459.95271
 Protein Bot Area: 1665.97419
 X Extent: -31 to 23
 Y Extent: -24 to 26
 Z Extent: -33 to 32
 Pore Minimum Radius: 0

Protein Projection onto XY:

Projection of protein portions above and below the membrane surface onto the membrane surface can be performed only to exclude pseudo lipid spheres underneath the protein portion during the lipid sphere packing.

- ☐ Turn on protein projection onto XY of upper leaflet
☐ Turn on protein projection onto XY of lower leaflet
 This option is not recommended to use unless it is absolutely necessary

System Size Determination Options:

☐ Homogeneous Lipid "Homogeneous Lipid" option is no longer supported. You can use "Heterogeneous Lipid" option even for homogeneous lipid bilayer building.

Heterogeneous Lipid

1. Box Type: Rectangular (Currently, only CHARMM, NAMD, and GROMACS support the hexagonal box)
2. Length of Z based on:

Water thickness 15 (Minimum water height on top and bottom of the system)

Hydration number 50 (Number of water molecules per one lipid molecule)
3. Length of XY based on:

Ratios of lipid components

Numbers of lipid components

Length of X and Y: 95 (initial guess)
(The system size along the X and Y must be the same)

Show the system info click this once you fill the following table:

Lipid Type	Charge [e]	Tail Info. [sn1/sn2]	Images	Upperleaflet Ratio (Integer)	Lowerleaflet Ratio (Integer)	Surface Area
Sterols						
PA (phosphatidic acid) Lipids						
PC (phosphatidylcholine) Lipids						
DDPC	0	10:0 / 10:0	[Image]	0	0	64.1
DCPC	0	11:0 / 11:0	[Image]	0	0	64.1
DLPC	0	12:0 / 12:0	[Image]	0	0	64.1
DMPC	0	14:0 / 14:0	[Image]	0	0	64.1
DPPC	0	16:0 / 16:0	[Image]	0	0	63.0
DSPC	0	18:0 / 18:0	[Image]	0	0	65.6
PSPC	0	16:0 / 18:0	[Image]	0	0	68.3
PYPC	0	16:0 / 16:1	[Image]	0	0	64.1
POPC	0	16:0 / 18:1	[Image]	1	1	68.3
PLPC	0	16:0 / 18:2	[Image]	0	0	65.2
PILPC	0	16:0 / 18:2	[Image]	0	0	60.0
PEPC	0	16:0 / 22:1	[Image]	0	0	66.5
SOPC	0	18:0 / 18:1	[Image]	0	0	66.2
SLPC	0	18:0 / 18:2	[Image]	0	0	65.2
DRPC	0	14:1 / 14:1	[Image]	0	0	67.0
DYPC	0	16:1 / 16:1	[Image]	0	0	67.0
YOPC	0	16:1 / 18:1	[Image]	0	0	67.0
DOPC	0	18:1 / 18:1	[Image]	0	0	69.7
OEPC	0	18:1 / 22:1	[Image]	0	0	66.5
DUPC	0	18:2 / 18:2	[Image]	0	0	70.0
DGPC	0	20:1 / 20:1	[Image]	0	0	66.7
DEPC	0	22:1 / 22:1	[Image]	0	0	66.5
DNPC	0	24:1 / 24:1	[Image]	0	0	62.3
DLIPC	0	18:2 / 18:2	[Image]	0	0	65.2
PE (phosphatidylethanolamine) Lipids						
PG (phosphatidylglycerol) Lipids						
PS (phosphatidylserine) Lipids						
PP (pyrophosphate) Lipids						
PI (phosphatidylinositol) Lipids						
CL (cardiolipin) Lipids						
DAG (diacylglycerol) Lipids ☐ Optimized charges for DAG						
TAG (triacylglycerol) Lipids ☐ Optimized charges for TAG						
TAP (trimethylammonium propane) & DAP/DMA (dimethylamino) Lipids						
PUFA (polyunsaturated fatty acid) Esterified Lipids						
SM (sphingo) and Ceramide Lipids						
Ether/Plasmalogen Lipids						
Bacterial Lipids						
Endosomal Lipids						
Ubiquinone/Ubiquinol & Archaeal Lipids						
Oxidized Lipids 2						

Calculated Number of Lipids:		
Lipid Type	Upperleaflet Number	Lowerleaflet Number
POPC	111	108

Calculated XY System Size:		
	Upperleaflet	Lowerleaflet
Protein Area	1459.95271	1665.97419
Lipid Area	7581.3	7376.4
# of Lipids	111	108
Total Area	9041.25271	9042.37419
Protein X Extent	30.20	
Protein Y Extent	25.70	
Average Area	9041.81	
A	95.09	
B	95.09	

- ▶ **Fatty Acids**
 - ▶ **Detergents**
 - ▶ **N-acylated Amino Acids**
 - ▶ **PEG (polyethylene glycol) Lipids**
 - ▶ **Glycolipids**
 - ▶ **LPS (lipopolysaccharides)**
-

Note: The surface areas listed above are only suggested values. In the case of high cholesterol concentration, 10~20% smaller surface areas will be adopted as default values, which might be more reasonable and close to equilibrium lipid areas. Please refer the manuscript or the following link to list of default surface area we are using. 

Next Step: 
Determine the System Size



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